

that the Windows-based Lumia still remains its flagship device, and the Nokia X platform will remain between its Lumia and Asha devices, with the Asha devices being priced even lower than the X series smartphones.

The newly launched device has 512 MB of RAM with a 1 GHz dual-core processor and 4 GB of internal storage, which can be taken up to 32 GB using a micro-SD card. In addition, it has a 1,500 mAh battery along with a 3 mega pixel fixed-focus camera.

With a price tag of Rs 8,599, some might say that the Nokia X is a bit overpriced for the features that it offers. In fact, the company's cheaper Lumia 520, though running on Windows, offers a better camera. This just reinforces the belief that Nokia is using this platform as a launch pad for its Lumia devices. Experts all over the world have reiterated over and over again that the X, X+ and XL are examples of Microsoft leveraging its own services using Google's Android.

The platform also grants users the option to run apps from third party app stores. In addition, Shroff mentioned during the product demonstration that users can side load

"Nokia has replaced things like Google Maps, Google Play and Google Drive with its own Here Maps, Nokia Store and Microsoft OneDrive. In addition, the user interface in this device looks almost exactly like the Windows Phone platform."

apk files for various apps onto this device. That said, there is no guarantee for the compatibility of side loaded apps and, hence, you have to ensure that they can run on the device.

So, that's the story of Nokia's new Android initiative. The one-time leader of the mobile phones market has added yet another ecosystem for users to experience. For developers, the Nokia X provides a new ecosystem to monetise, but whether the platform will be successful enough remains to be seen. 

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Variable eliminations and equation solving

Very often, we come across 'N' sets of equations in 'M' sets of unknowns, where $M \geq N$. If $M = N$, then it's most likely that a unique solution exists. However, if $M > N$, then there may be many possible solutions, but with some constraints. In such a case, it would be helpful to deduce some simpler set of expressions. This can be achieved using `eliminate()` of Maxima. Let's have two polynomial expressions in the three variables x , y and z to demonstrate this:

```
$ maxima -q
(%i1) p1: x^2 + 2*x*y + z^2$
(%i2) p2: x + y + z$
(%i3) string(eliminate([p1, p2], [x]));
(%o3) [2*z^2-y^2]
(%i4) string(eliminate([p1, p2], [y]));
(%o4) [-z^2+2*x*z+x^2]
(%i5) string(eliminate([p1, p2], [z]));
(%o5) [y^2+4*x*y+z^2]
(%i6) quit();
```

Note that in all the above polynomial expressions, the expressions are assumed to be equated to zero. A beautiful application of the above is solving recurrence relations. Let's assume that we have a non-linear equation given by $Y_{n+1} = r * Y_n * (1 - Y_n)$. And, we want to solve it for a scenario where as 'n' tends to infinity, the value of 'Y' oscillates between two distinct values. This means that $Y_{n+2} = Y_n$. Hence, we would have two expressions with three unknowns to solve, namely:

$$1) Y_{n+1} = r * Y_n * (1 - Y_n)$$

$$2) Y_n = r * Y_{n+1} * (1 - Y_{n+1})$$

So, representing Y_{n+1} with $yn1$ and Y_n by yn , we may use `solve()` to solve for yn and $yn1$ in terms of r . But, if we want to get a feel of the equation which pops up, in terms of yn , we would have to use `eliminate()` to eliminate $yn1$.

```
$ maxima -q
(%i1) p1: yn1 = r * yn * (1 - yn)$
(%i2) p2: yn = r * yn1 * (1 - yn1)$
(%i3) string(eliminate([p1, p2], [yn1]));
(%o3) [yn*(r^3*yn^3-2*r^3*yn^2+(r^3+r^2)*yn-r^2+1)]
(%i4) string(solve([p1, p2], [yn, yn1])); /* Just to show the solution */
(%o4) [[yn = (r-1)/r, yn1 = (r-1)/r], [yn = -(sqrt(r^2-2*r-3)-r-1)/(2*r), yn1 = (sqrt(r-3)*sqrt(r+1)+r+1)/(2*r)], [yn = (sqrt(r^2-2*r-3)+r+1)/(2*r), yn1 = -(sqrt(r-3)*sqrt(r+1)-r-1)/(2*r)], [yn = 0, yn1 = 0]]
(%i5) quit();
```



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